



Chapter 5

Discriminant Analysis



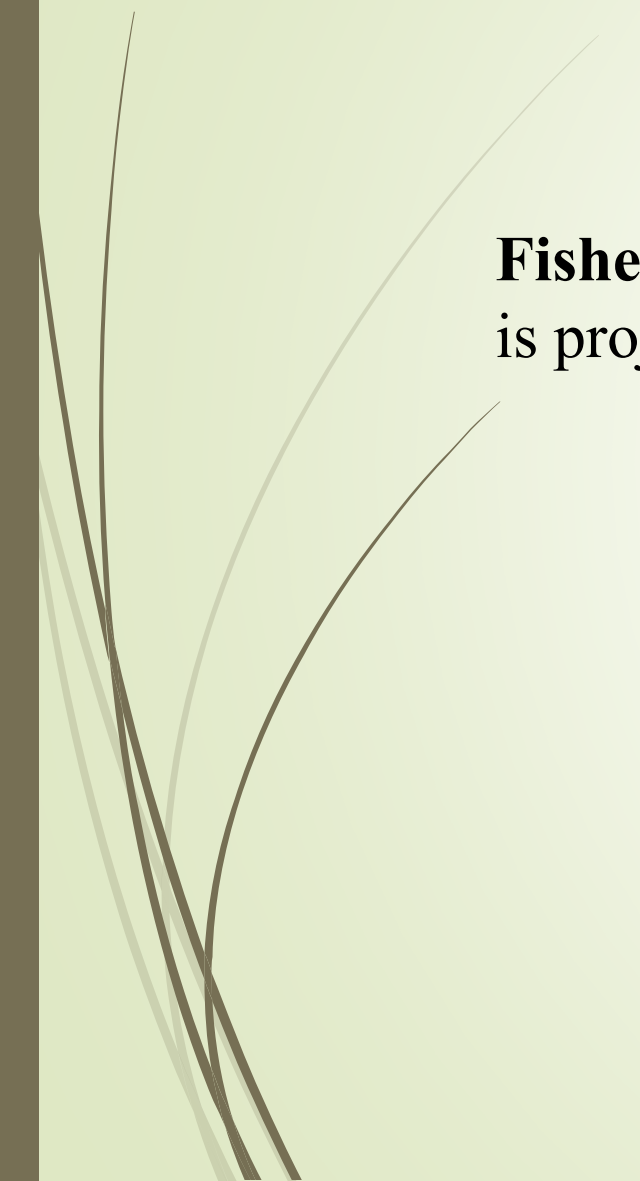
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§ 5.4 Fisher Discriminant

Fisher Discriminant(费希尔判别) : Its basic idea is projection (or dimensionality reduction):



§ 5.4 Fisher Discriminant

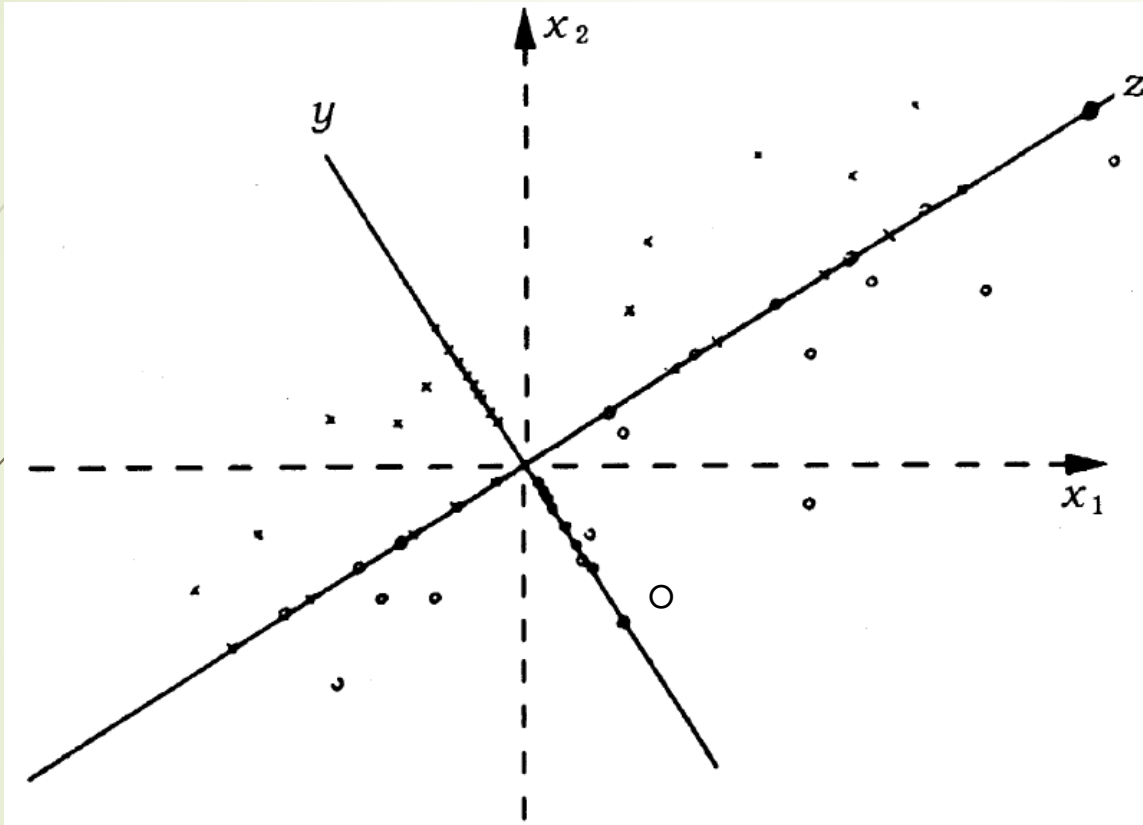
- **Fisher Discriminant(费希尔判别)** : Its basic idea is projection (or dimensionality reduction): Use several linear combinations of p – vector $\mathbf{x} = (x_1, x_2, \dots, x_p)'$ (called Discriminant (判别式))

$$y_1 = \mathbf{a}'_1 \mathbf{x}, y_2 = \mathbf{a}'_2 \mathbf{x}, \dots, y_r = \mathbf{a}'_r \mathbf{x}$$

(usually, r is much smaller than p) to replace original p variables $x_1, x_2, \dots, x_p$ to achieve dimension reduction. Discriminate which group samples $x_1, x_2, \dots, x_p$ belongs to based on discriminants, $y_1, y_2, \dots, y_r$. Successful dimension reduction makes discrimination more convenient and efficient. In addition, we can draw figures of the former 2 or 3 discriminants and make discrimination from intuitive geometric figures.

§ 5.4 Fisher Discriminant

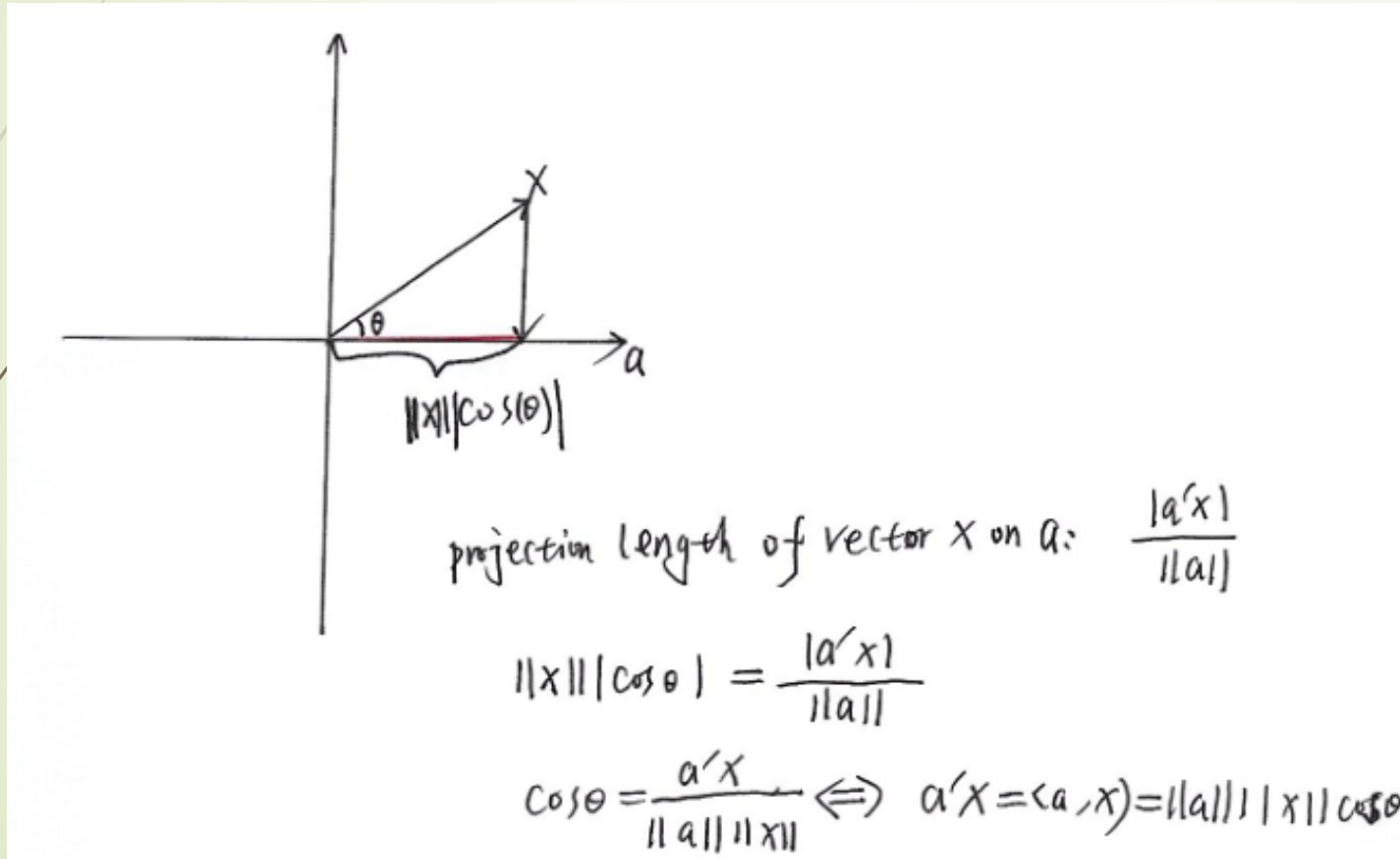
Two-dimensional example



- Consider project k groups of p -vectors to a certain optimal direction in space R^p , a , that is to say, the projection points on a can reflect the difference among groups at utmost.

§ 5.4 Fisher Discriminant

- Projection: projection length of vector x on a : $|a'x|/\|a\|$, projection value is $a'x/\|a\|$.



► For Fisher discriminant, we need to assume $\Sigma_1 = \Sigma_2 = \dots = \Sigma_k = \Sigma$. Suppose $\mathbf{x}_{ij}$ are p -dimensional observed value from $\pi_i, j=1,2,\dots,n_i, i=1,2,\dots,k$. Project them to a certain p -dimensional constant variable, a . The projection points will be $y_{ij} = a' \mathbf{x}_{ij}$.

► Denote $\bar{\mathbf{x}}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} \mathbf{x}_{ij}, \quad \bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^k n_i \bar{\mathbf{x}}_i, \quad n = \sum_{i=1}^k n_i$
 $\bar{y}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} y_{ij} = a' \bar{\mathbf{x}}_i, \quad \bar{y} = \frac{1}{n} \sum_{i=1}^k \sum_{j=1}^{n_i} y_{ij} = \frac{1}{n} \sum_{i=1}^k n_i \bar{y}_i = a' \bar{\mathbf{x}},$

Total regression sum of squares(回归平方) for the projection points y_{ij}

$$SSTR = \sum_{i=1}^k n_i (\bar{y}_i - \bar{y})^2 = \sum_{i=1}^k n_i (a' \bar{\mathbf{x}}_i - a' \bar{\mathbf{x}})^2 = a' B a.$$

where $B = \sum_{i=1}^k n_i (\bar{\mathbf{x}}_i - \bar{\mathbf{x}})(\bar{\mathbf{x}}_i - \bar{\mathbf{x}})'$ is the cross product matrix of total regression sum of squares(组间平方和及叉积矩阵).

- The projection points $y_{ij}, j=1, 2, \dots, n_i, i=1, 2, \dots, k$. The error sum of squares:

$$SSE = \sum_{i=1}^k \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_i)^2 = \sum_{i=1}^k \sum_{j=1}^{n_i} (a' x_{ij} - a' \bar{x}_i)^2 = a' E a$$

$$E = \sum_{i=1}^k (n_i - 1) S_i = \sum_{i=1}^k \sum_{j=1}^{n_i} (\mathbf{x}_{ij} - \bar{\mathbf{x}}_i)(\mathbf{x}_{ij} - \bar{\mathbf{x}}_i)' \quad \text{组内平方和及叉积矩阵}$$

- SSTR and SSE are respectively with free degree $k-1$ and $n-k$. In univariate covariance analysis(P113), F statistic is defined as:

$$F = \frac{SSTR / (k - 1)}{SSE / (n - k)} = \frac{a' B a / (k - 1)}{a' E a / (n - k)} \quad (4.1.25)$$

is distributed as $F(k - 1, n - k)$. The larger the F -value, the easier it is to distinguish between the means of the k groups.

- We should choose such a direction vector a (satisfying $a' S_p a = 1, S_p = E / (n - k)$) that maximizes the value of the above F -statistic, that is to make

$$\Delta(a) = \frac{a' B a}{a' E a} \quad (5.4.1)$$

attain the maximum. From(1.8.2), the maximum of $\Delta(a)$ is the largest eigenvalue of $E^{-1} B$.

§ 1.8 Extremal Problem of Eigenvalue

- ➡ (1) If A is p dimensional symmetric matrix and its eigenvalues are $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p$, then

$$\max_{x \neq 0} \mathbf{x}' A \mathbf{x} / \mathbf{x}' \mathbf{x} = \lambda_1$$

$$\min_{x \neq 0} \mathbf{x}' A \mathbf{x} / \mathbf{x}' \mathbf{x} = \lambda_p$$

- ➡ (2) If A is p dimensional symmetric matrix, B is p dimensional positive definite matrix and $\mu_1 \geq \mu_2 \geq \cdots \geq \mu_p$ are the eigenvalues of $B^{-1}A$, then

$$\max_{x \neq 0} \mathbf{x}' A \mathbf{x} / \mathbf{x}' B \mathbf{x} = \mu_1$$

$$\min_{x \neq 0} \mathbf{x}' A \mathbf{x} / \mathbf{x}' B \mathbf{x} = \mu_p$$

- ➡ (3) (Cauchy–Schwarz Inequation) 若 $B > 0$, 则

$$(\mathbf{x}' \mathbf{y})^2 \leq (\mathbf{x}' B \mathbf{x})(\mathbf{y}' B^{-1} \mathbf{y})$$

- Suppose all the non-zero eigenvalues of $E^{-1}B$ are successively $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_s > 0$, where $s = \text{rank}(B) \leq \min(k-1, p)$ (exercise 5.6). Corresponding eigenvectors are respectively $t_1, t_2, \dots, t_s$ (standardized as $t_i' S_p t_i = 1, i=1, 2, \dots, s$) satisfying $(B - \lambda_i E)t_i = 0, i = 1, 2, \dots, s$

- Plug $t_1, t_2, \dots, t_s$ into (5.4.1):

$$\Delta(t_i) = \frac{t_i' B t_i}{t_i' E t_i} = \frac{t_i' (\lambda_i E t_i)}{t_i' E t_i} = \lambda_i,$$

- $y_1 = t_1' x$ is called First Discriminant(第一判别式), $y_2 = t_2' x$ is called Second Discriminant(第二判别式). Generally, $y_i = t_i' x$ is called i th Discriminant(第*i*判别式), $i=1, 2, \dots, s$.
- Eigenvalue λ_i reflects the contribution of i -th discriminant y_i to discriminating groups. Contribution rate of y_i

$$\lambda_i / \sum_{i=1}^s \lambda_i$$



the accumulative contribution rate of the former $r(\leq s)$ discriminants $y_1, y_2, \dots, y_r$

$$\sum_{i=1}^r \lambda_i / \sum_{i=1}^s \lambda_i$$

which reflects the discrimination ability of $y_1, y_2, \dots, y_r$.

- ➡ In real application, if the accumulative contribution rate of the former $r(\leq s)$ discriminants has reached to a higher proportion (like 75%~95%), then the r discriminants can be used to discriminate.

For new sample $\mathbf{x} = (x_1, x_2, \dots, x_p)'$, $y_1 = t'_1 \mathbf{x}$, $y_2 = t'_2 \mathbf{x}$, $\dots$, $y_r = t'_r \mathbf{x}$ are the first r Discriminants. By using the above r discriminants, we obtain

Fisher Discrimination rule as follows:

$$\mathbf{x} \in \pi_l, \quad \text{if } \sum_{j=1}^r (y_j - \bar{y}_{lj})^2 = \min_{1 \leq i \leq k} \sum_{j=1}^r (y_j - \bar{y}_{ij})^2$$

where $\bar{y}_{ij}$ is the sample mean of the j -th discriminant in group π_i and

$$\bar{y}_{ij} = \frac{1}{n_i} \sum_{j=1}^{n_i} y_{ij} = \frac{1}{n_i} \sum_{j=1}^{n_i} t'_j x_{ij} = t'_j \bar{x}_i, \quad \bar{x}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} x_{ij}, \quad i = 1, 2, \dots, k.$$

The discrimination rule can also be shown as follows:

$$\mathbf{x} \in \pi_l, \quad \text{if } \sum_{j=1}^r [t'_j (\mathbf{x} - \bar{\mathbf{x}}_l)]^2 = \min_{1 \leq i \leq k} \sum_{j=1}^r [t'_j (\mathbf{x} - \bar{\mathbf{x}}_i)]^2 \quad (5.4.6)$$

- If only one discriminant is used to discriminate ($r=1$), then the above discrimination rule can be simplified as:

$$\mathbf{x} \in \pi_l, \quad \text{if } |y - \bar{y}_l| = \min_{1 \leq i \leq k} |y - \bar{y}_i|$$

y and $\bar{y}_i (i=1,2,\dots,k)$ are respectively y_1 and $\bar{y}_{i1} (i=1,2,\dots,k)$ in the former rule.

- Sometimes, we also use **Centralized Fisher Discriminant**(中心化的费希尔判别式), i.e.,:

$$y_i = \mathbf{t}'_i (\mathbf{x} - \bar{\mathbf{x}}), \quad i = 1, 2, \dots, s$$

$$\bar{y}_{ij} = \mathbf{t}'_j (\bar{\mathbf{x}}_i - \bar{\mathbf{x}}), \quad j = 1, 2, \dots, n_i$$

$\bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^k \sum_{j=1}^{n_i} \mathbf{x}_{ij}$ is the total mean of k groups. Still use the rule (5.4.6) to discriminate.

$$\mathbf{x} \in \pi_l, \quad \text{if } \sum_{j=1}^r [\mathbf{t}'_j (\mathbf{x} - \bar{\mathbf{x}}_l)]^2 = \min_{1 \leq i \leq k} \sum_{j=1}^r [\mathbf{t}'_j (\mathbf{x} - \bar{\mathbf{x}}_i)]^2 \quad (5.4.6)$$

Example 5.4.2

- Iris(鸢尾花) Data published by Fisher in 1936 are widely used as the discriminant analysis example. Data were collected for 3 kinds of Iris with each sample size 50 :bristles (刚毛) iris (I), discoloration (变色) iris(II) and Virginia(弗吉亚)iris(III). These data including calyx(花萼) length(x_1), calyx(花萼) width(x_2), petal(花瓣) length(x_3), petal (花瓣)width(x_4), with mm unit and are listed in Table 5.4.1.

Table 5.4.1

Iris Data

Number	Group	x_1	x_2	x_3	x_4	Number	Group	x_1	x_2	x_3	x_4
1	I	50	33	14	2	76	III	58	27	51	19
2	III	64	28	56	22	77	II	57	29	42	13
3	II	65	28	46	15	78	III	72	30	58	16
4	III	67	31	56	24	79	I	54	34	15	4
5	III	63	28	51	15	80	I	52	41	15	1
6	I	46	34	14	3	81	III	71	30	59	21
7	III	69	31	51	23	82	III	64	31	55	18
8	II	62	22	45	15	83	III	60	30	48	18
9	II	59	32	48	18	84	III	63	29	56	18
10	I	46	36	10	2	85	II	49	24	33	10
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
66	II	56	30	45	15	141	II	55	23	40	13
67	II	58	27	41	10	142	II	66	30	44	14
68	I	50	34	16	4	143	II	68	28	48	14
69	I	46	32	14	2	144	I	54	34	17	2
70	II	60	29	45	15	145	I	51	37	15	4
71	II	57	26	35	10	146	I	52	35	15	2
72	I	57	44	15	4	147	III	58	28	51	24
73	I	50	36	14	2	148	II	67	30	50	17
74	III	77	30	61	23	149	III	63	33	60	25
75	III	63	34	56	24	150	I	53	37	15	2

➡ $n_1=n_2=n_3=50$, $n=n_1+n_2+n_3=150$. After calculation,

$$\bar{\mathbf{x}}_1 = \begin{pmatrix} 50.06 \\ 34.28 \\ 14.62 \\ 2.46 \end{pmatrix}, \quad \bar{\mathbf{x}}_2 = \begin{pmatrix} 59.36 \\ 27.70 \\ 42.60 \\ 13.26 \end{pmatrix}, \quad \bar{\mathbf{x}}_3 = \begin{pmatrix} 65.88 \\ 29.74 \\ 55.52 \\ 20.26 \end{pmatrix}$$

$$\bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^3 n_i \bar{\mathbf{x}}_i = \begin{pmatrix} 58.433 \\ 30.573 \\ 37.580 \\ 11.993 \end{pmatrix}$$



$$B = \sum_{i=1}^3 n_i (\bar{\mathbf{x}}_i - \bar{\mathbf{x}})(\bar{\mathbf{x}}_i - \bar{\mathbf{x}})'$$

$$= \begin{pmatrix} 6321.213 & -1995.267 & 16524.840 & 7127.933 \\ -1995.267 & 1134.493 & -5723.960 & -2293.267 \\ 16524.840 & -5723.960 & 43710.280 & 18677.400 \\ 7127.933 & -2293.267 & 18677.400 & 8041.333 \end{pmatrix}$$

$$E = \sum_{i=1}^3 \sum_{j=1}^{n_i} (\bar{\mathbf{x}}_{ij} - \bar{\mathbf{x}}_i)(\bar{\mathbf{x}}_{ij} - \bar{\mathbf{x}}_i)'$$

$$= \begin{pmatrix} 3895.620 & 1363.000 & 2462.460 & 564.500 \\ 1363.000 & 1696.200 & 812.080 & 480.840 \\ 2462.460 & 812.080 & 2722.260 & 627.180 \\ 564.500 & 480.840 & 627.180 & 615.660 \end{pmatrix}$$


$$\mathbf{E}^{-1}\mathbf{B} = \begin{pmatrix} -3.058 & 1.081 & -8.112 & -3.459 \\ -5.562 & 2.178 & -14.965 & -6.308 \\ 8.077 & -2.943 & 21.512 & 9.142 \\ 10.497 & -3.420 & 27.549 & 11.846 \end{pmatrix}$$

The number of positive eigenvalues of $\mathbf{E}^{-1}\mathbf{B}$ is $s \leq \min(k-1, p) = \min(2, 4) = 2$,
the two positive eigenvalues of $\mathbf{E}^{-1}\mathbf{B}$ are:

$$\lambda_1 = 32.192, \quad \lambda_2 = 0.285,$$

and the corresponding standardized eigenvectors are:

$$\mathbf{t}_1 = \begin{pmatrix} -0.083 \\ -0.153 \\ 0.220 \\ 0.281 \end{pmatrix}, \quad \mathbf{t}_2 = \begin{pmatrix} 0.002 \\ 0.216 \\ -0.093 \\ 0.284 \end{pmatrix}$$

Therefore, centralized Fisher discriminants are:

$$\begin{aligned}y_1 &= \mathbf{t}'_1 (\mathbf{x} - \bar{\mathbf{x}}) \\&= -0.083 \times (x_1 - 58.433) - 0.153 \times (x_2 - 30.573) \\&\quad + 0.220 \times (x_3 - 37.580) + 0.281 \times (x_4 - 11.993) \\y_2 &= \mathbf{t}'_2 (\mathbf{x} - \bar{\mathbf{x}}) \\&= 0.002 \times (x_1 - 58.433) + 0.216 \times (x_2 - 30.573) \\&\quad - 0.093 \times (x_3 - 37.580) + 0.284 \times (x_4 - 11.993)\end{aligned}$$

Group means of discriminants:

$$\begin{aligned}\bar{y}_{11} &= -7.608, & \bar{y}_{21} &= 1.825, & \bar{y}_{31} &= 5.783 \\ \bar{y}_{12} &= 0.215, & \bar{y}_{22} &= -0.728, & \bar{y}_{32} &= 0.513\end{aligned} \quad \bar{y}_{ij} = \mathbf{t}'_j \bar{\mathbf{x}}_i$$

For any sample $\mathbf{x}$, it can be discriminated as the following :

$$\mathbf{x} \in \pi_l, \quad \text{if } \sum_{j=1}^2 (y_j - \bar{y}_{lj})^2 = \min_{1 \leq i \leq 3} \sum_{j=1}^2 (y_j - \bar{y}_{ij})^2$$

- Since n_1, n_2, n_3 are big, the result of estimating misclassification probability using (5.2.7) is good. Discrimination results are listed in Table 5.4.2.

Table 5.4.2

Discrimination Result

discrimination				
facts		I	II	III
	I	50	0	0
	II	0	48	2
	III	0	1	49

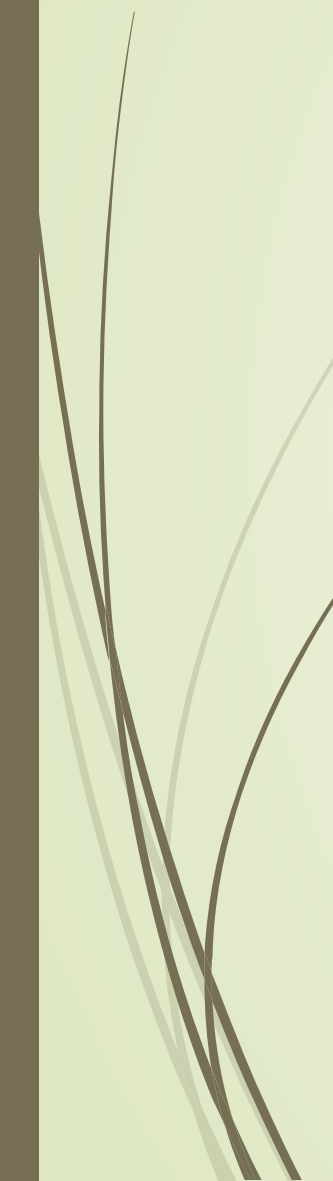
So

$$\hat{P}(2|1) = 0, \quad \hat{P}(3|1) = 0$$

$$\hat{P}(1|2) = 0, \quad \hat{P}(3|2) = \frac{2}{50} = 0.04$$

$$\hat{P}(1|3) = 0, \quad \hat{P}(2|3) = \frac{1}{50} = 0.02$$

these misclassification probabilities are relatively low.

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- We can draw a scatter plot using 150 sample discriminants(y_1, y_2). The following figure is the output of SAS9.1. In the figure, Can1, Can2 are respectively y_1, y_2 . Points from I, II and III are scaled out as “1”, “2” and “3”.
 - According to the plot, separated results are excellent. For a new sample x_0 , we can use ocular estimate method(目测法) to discriminate its group intuitively. We need to point out that ocular estimate method is the principal value of Fisher discriminant. Plots usually can reflect the plentiful information that computers cannot thereby effective discrimination.

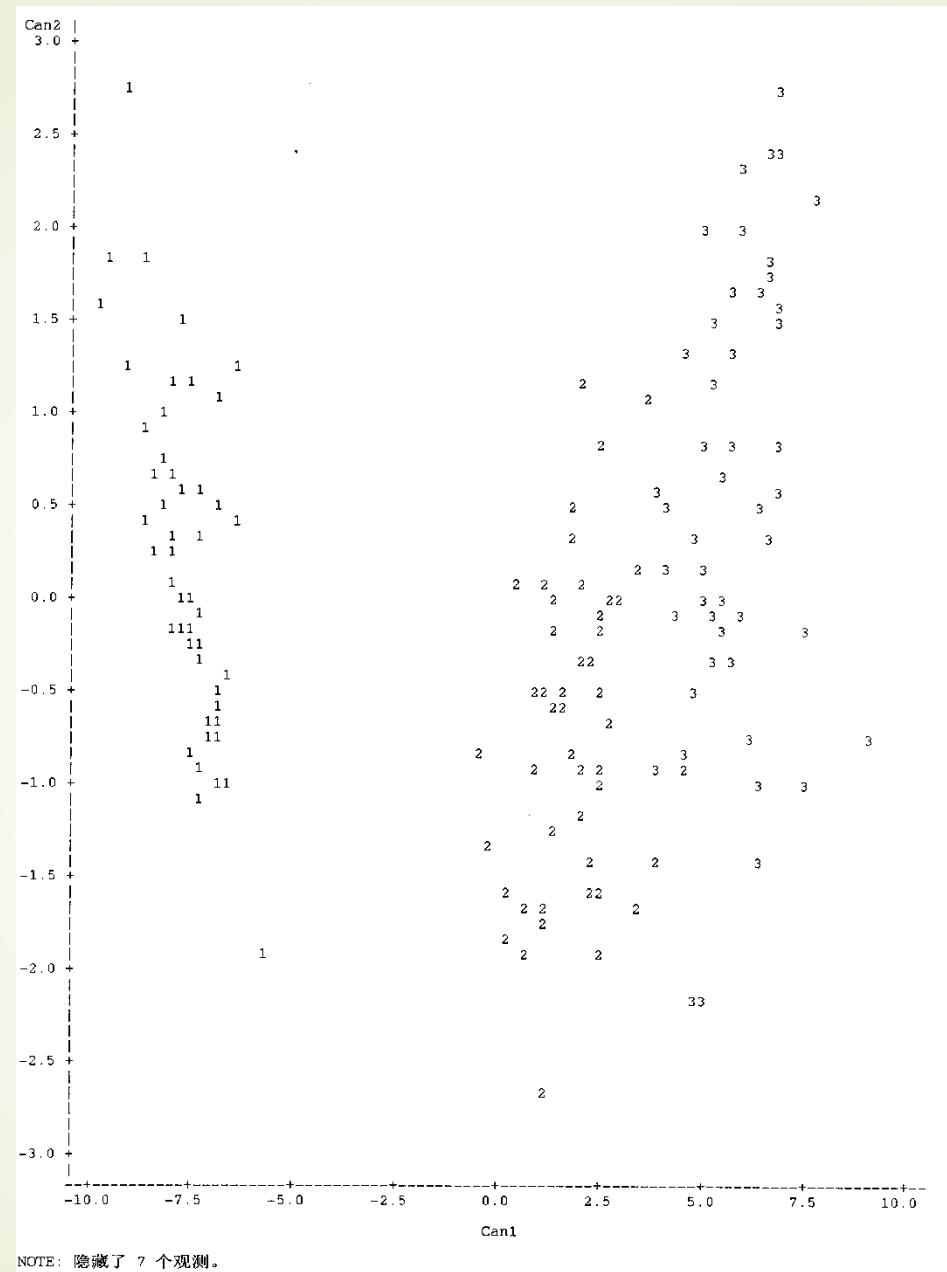


Figure 5.4.2 Scatter Plot for 2 discriminants



Assignment: See the assignment file



§ 5.5 Stepwise Discrimination

- In this chapter, we discussed how to perform discriminant analysis based on observed variable data. However, an equally important question is: which variables should be selected for the discriminant analysis. In practice, a large number of variables are often initially chosen based on domain knowledge and expert judgment. In such cases, we hope to choose a subset of variables that contains fewer variables while retaining almost as much information as the original variable set. This is the goal of the method introduced in this section, which is actually a variable selection problem in the context of discriminant analysis.
- Stepwise discriminant analysis(逐步判别法) is an automatic variable subset selection method used in discriminant analysis. While it may not always yield the optimal solution, it is often effective in practice and remains one of the most widely used techniques for selecting discriminant variables.

§ 5.4 Stepwise Discrimination

I), The test of additional information(附加信息检验)

- Let $x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$, where $x_1 = (x_1, x_2, \dots, x_r)'$ represents the original variables used for discrimination, and $x_2 = (x_{r+1}, x_{r+2}, \dots, x_p)'$ represents the newly introduced variables. We want to know whether, given x_1 as the discriminant variables, the additional information provided by x_2 can significantly improve the ability to distinguish groups. If there is no significant improvement, then the introduction of x_2 is deemed unworthy.
- Suppose there are k groups $\pi_1, \pi_2, \dots, \pi_k$, all follow a p -dimensional normal distribution with the same covariance matrix. Samples are independently drawn from these k groups. Let $x_{i1}, x_{i2}, \dots, x_{in_i}$ denote the samples taken from population π_i , where n is the total sample size for these k groups. The above problem leads to the following hypothesis testing problem:

$$H_0: E(x_2|x_1) \text{ are equal across all groups,} \quad (5.5.1)$$

$$H_1: E(x_2|x_1) \text{ are not equal at least for two groups}$$

- This is a test to determine whether x_2 has a significant effect on distinguishing between groups under the condition that x_1 is selected.

§ 5.5 Stepwise Discrimination

■ I) Notations:

sum of error squares and cross product matrix (组内(误差)平方和及叉积和矩阵(简称组内矩阵))

$$E = SSE = \sum_{i=1}^k \sum_{j=1}^{n_i} (x_{ij} - \bar{x})(x_{ij} - \bar{x})'$$

sum of regression squares and cross product matrix (组间平方和及叉积和矩阵(简称组间矩阵))

$$H = SSTR = \sum_{i=1}^k n_i (\bar{x}_i - \bar{x})(\bar{x}_i - \bar{x})' \longrightarrow df = k - 1$$

E and H are divided into

$$E = \begin{bmatrix} E_{11} & E_{12} \\ E_{21} & E_{22} \end{bmatrix}, H = \begin{bmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{bmatrix},$$

where E_{11} , H_{11} are the $r \times r$ matrix, E_{22} , H_{22} are the $(p-r) \times (p-r)$ matrix. The test statistic for the hypothesis (5.5.1) is

$$(5.5.2) \quad \Lambda(x_2|x_1) = \frac{\Lambda(x_1, x_2)}{\Lambda(x_1)},$$

$$\text{where, } \Lambda(x_1, x_2) = \frac{|E|}{|E+H|}, \Lambda(x_1) = \frac{|E_{11}|}{|E_{11}+H_{11}|}.$$

Using likelihood ratio method, we can obtain Wilks Λ Statistic (威尔克斯统计量)

$$\Lambda_{p,k-1,n-k} = \frac{|SSE|}{|SSE + SSTR|} = \frac{|SSE|}{|SST|}$$

with $\Lambda_{p,k-1,n-k} \in [0,1]$.

$$(4.5.6)$$

When H_0 is true, Wilks Λ Statistic $\Lambda(x_2|x_1)$ follows $\Lambda(p-r, k-1, n-k-r)$.

§ 5.5 Stepwise Discrimination

- 1), The test statistic for the hypothesis (5.5.1) is

$$\Lambda(x_2|x_1) = \frac{\Lambda(x_1, x_2)}{\Lambda(x_1)}, \quad (5.5.2)$$

where, $\Lambda(x_1, x_2) = \frac{|E|}{|E+H|}$, $\Lambda(x_1) = \frac{|E_{11}|}{|E_{11}+H_{11}|}$.

When H_0 is true, Wilks Λ Statistic $\Lambda(x_2|x_1)$ follows $\Lambda(p-r, k-1, n-k-r)$.

- The properties for $\Lambda \sim \Lambda(p, n, m)$:

(1) $\Lambda(p, n, m) = \Lambda(n, p, m+n-p)$; (4-3-2)

(2) When $p=2$, $\frac{(m-1)(1-\sqrt{\Lambda})}{n\sqrt{\Lambda}} \sim F(2n, 2(m-1))$; (4-3-3)

(3) When $p=1$, $\frac{m(1-\Lambda)}{n\Lambda} \sim F(n, m)$. (4-3-5)

§ 5.5 Stepwise Discrimination

➤ When $p - r = 1$, it follows that

$$\Lambda = \Lambda(x_p | x_1, x_2, \dots, x_{p-1}) = \frac{\Lambda(x_1, x_2, \dots, x_p)}{\Lambda(x_1, x_2, \dots, x_{p-1})} \quad (5.5.3)$$

is distributed as $\Lambda(1, k - 1, n - k - p + 1)$. The statistic in (5.5.3) is commonly referred to as **the partial Λ statistic**. From (4-3-5), we have $\Lambda \sim \Lambda(1, n, m), \frac{m(1-\Lambda)}{n\Lambda} \sim F(n, m)$, (4-3-5), and

$$F = F(x_p | x_1, x_2, \dots, x_{p-1}) = \frac{n-k-p+1}{k-1} \frac{1-\Lambda}{\Lambda}, \quad (5.5.4)$$

which is distributed as $F(k - 1, n - k - p + 1)$. The statistic in (5.5.4) is common referred to as **the partial F statistic**.

For a given significance level α , the rejection rule is:

$$\text{If } F \geq F_\alpha(k - 1, n - k - p + 1), \text{ then reject } H_0. \quad (5.5.5)$$

§ 5.5 Stepwise Discrimination

二、变量选择的方法

给定显著水平 α ，逐步判别法的基本步骤如下：

1) 对于每个 x_i ，计算其一元方差分析的F统计量 $F(x_i)$ ，不妨设 $F(x_1)=\max_i F(x_i)$ ，

即 x_1 有最大的判别能力。若 $F(x_1) < F_\alpha(k-1, n-k)$ ，则表明没有一个变量可以选入；
若 $F(x_1) \geq F_\alpha(k-1, n-k)$ ，则 x_1 选入，并进入下一步。

2) 对1) 每一未选变量，计算偏F统计量 $F(x_i|x_1)$ ，不妨设 $F(x_2|x_1)=\max_{2 \leq i \leq p} F(x_i|x_1)$ ，

即 x_2 对判别能力的提升有最大的贡献。若 $F(x_2|x_1) < F_\alpha(k-1, n-k-1)$ ，则选变量过程结束；
若 $F(x_2|x_1) \geq F_\alpha(k-1, n-k-1)$ ，则 x_2 选入，并进入下一步。

一般地，如已选了 r 个变量，不妨设是 $x_1, x_2, \dots, x_r$ ，并设

$F(x_{r+1}|x_1, x_2, \dots, x_r) = \max_{r+1 \leq i \leq p} F(x_i|x_1, x_2, \dots, x_r)$ 。若

$F(x_{r+1}|x_1, \dots, x_r) < F_\alpha(k-1, n-k-r)$ ，则选变量过程结束；否则， x_{r+1} 选入，
并进入下一步

§ 5.5 Stepwise Discrimination

二、变量选择的方法

给定显著水平 α ，逐步判别法的基本步骤如下：

3) 在第 $r + 1$ 个变量选入后，要重新核实较早选入的 r 个变量，将对判别效果不再显著的变量剔除出去。不妨设

$$F(x_l|x_1, \dots, x_{l-1}, x_{l+1}, \dots, x_{r+1}) = \min_{1 \leq i \leq r} F(x_i|x_1, x_2, \dots, x_{i-1}, x_{i+1}, \dots, x_{r+1}),$$

若 $F(x_l|x_1, \dots, x_{l-1}, x_{l+1}, \dots, x_{r+1}) \geq F_\alpha(k-1, n-k-r)$, 则没有变量被剔除，回到2)，

若 $F(x_l|x_1, \dots, x_{l-1}, x_{l+1}, \dots, x_{r+1}) < F_\alpha(k-1, n-k-r)$, 则剔除变量 x_l ，再对其余 $r-1$ 个变量继续进行核实，直到无变量可删除为止，然后再回到2)。

4) 经过2) 和3) 不断地选入和剔除过程，最后既不能选进新变量，也不能剔除已选入的变量，变量选择过程结束。

进行逐步判别实际上是在做逐步多元方差分析，在变量的筛选过程中没有任何函数被计算

► 例5.5.1对例5.4.2中的数据（见Table 5.4.1）做逐步判别，具体步骤如下：

(1) 对每一变量分别计算一元方差分析的F统计量和p值，并列于表5.5.1.因 x_3 相应的F=1180.16为最大，且 $p<0.0001$ ，故 x_3 第一个选入，它是最能分离各组的变量

表5.5.1 F统计量和p值

变量	x_1	x_2	x_3	x_4
F	119.26	49.16	1180.16	960.01
p-value	<0.0001	<0.0001	<0.0001	<0.0001

$$SSTR = \sum_{i=1}^k n_i (\bar{x}_i - \bar{x})^2$$
$$SSE = \sum_{i=1}^k \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_i)^2$$
$$F = \frac{SSTR / (k - 1)}{SSE / (n - k)}$$

$$F = \frac{6321.213/(3-1)}{3895.62/(150-3)} = 119.26 \text{ (Refer to Example 4.6.1)}$$

(2) 计算偏F统计量 $F(x_i|x_3)$ 及p值，由(5.5.3)和(5.5.4)知，

$$\Lambda(x_i|x_3) = \frac{\Lambda(x_3, x_i)}{\Lambda(x_3)}$$

$$F = F(x_p|x_1, x_2, \dots, x_{p-1}) = \frac{n-k-p+1}{k-1} \frac{1-\Lambda}{\Lambda}, \quad (5.5.4)$$

$$F(x_i|x_3) = \frac{1-\Lambda(x_i|x_3)}{\Lambda(x_i|x_3)} \frac{n-k-1}{k-1}, i=1, 2, 4, \sim F(k-1, n-k-1) \text{ with } n=150, k=3$$

§ 5.5 Stepwise Discrimination

计算结果列于表5.5.2. 因 x_2 具有最大的偏F, 且 $p<0.0001$, 于是 x_2 选入

表5.5.2 x_3 选入时的偏 F统计量和p值

变量	x_1	x_2	x_4
偏 F	34.32	43.04	24.77
p-value	<0.0001	<0.0001	<0.0001

3) 核实 x_3 是否因 x_2 的选入仍保持显著。经计算, $F(x_3|x_2)=1112.95$, $p<0.0001$,从而保留 x_3 .接着仍按(5.5.3)和(5.5.4)式计算偏F统计量 $F(x_i|x_2, x_3)$,即有

$$\Lambda(x_i|x_2, x_3) = \frac{\Lambda(x_2, x_3, x_i)}{\Lambda(x_2, x_3)}$$

$$F(x_i|x_2, x_3) = \frac{1-\Lambda(x_i|x_2, x_3))}{\Lambda(x_i|x_2, x_3))} \frac{n-k-2}{k-1} \sim F(k-1, n-k-2), i=1, 4, \text{ 结果见表5.5.3. 可见 } x_4 \text{ 选入}$$

表5.5.3 x_2, x_3 选入时的偏 F统计量和p值

变量	x_1	x_4
偏 F	12.27	34.57
p-value	<0.0001	<0.0001

§ 5.5 Stepwise Discrimination

4) 核实 x_4 选入后早先已选入的 x_2 和 x_3 是否还显著。计算偏F统计量 $F(x_2|x_3, x_4)$ 和 $F(x_3|x_2, x_4)$ ，结果列于表5.5.4.可见， x_2 和 x_3 皆保留。继续计算

表5.5.4 选入 x_4 后核实 x_2 和 x_3 是否还显著的的偏 F统计量和p值

变量	x_2	x_3
偏 F	54.58	38.72
p-value	<0.0001	<0.0001

$$\Lambda(x_1|x_2, x_3, x_4) = \frac{\Lambda(x_1, x_2, x_3, x_4)}{\Lambda(x_2, x_3, x_4)}$$

$$F(x_1|x_2, x_3, x_4) = \frac{1 - \Lambda(x_1|x_2, x_3, x_4)}{\Lambda(x_1|x_2, x_3, x_4)} \frac{n - k - 3}{k - 1} \sim F(k - 1, n - k - 3).$$

可得 $F(x_1|x_2, x_3, x_4) = 4.72, p = 0.0103$, 故 x_1 也进入。

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5) 核实 x_1 选入后原已选入的 x_2 , x_3 和 x_4 是否还显著。计算偏F统计量

$F(x_2|x_1, x_3, x_4)$, $F(x_3|x_1, x_2, x_4)$ 和 $F(x_4|x_1, x_2, x_3)$, 结果列于表5.5.5. 计算结果表明, 已选入的变量无一删除.

表5.5.5 选入 x_1 后核实 x_2, x_3 和 x_4 是否还显著的偏 F统计量和p值

变量	x_2	x_3	x_4
偏F	21.94	35.59	24.90
p-value	<0.0001	<0.0001	<0.0001

6)我们将上述变量选择过程汇总于表5.5.6中。

表5.5.6 变量选择过程汇总

步骤	1	2	3	4
变量	x_3	x_2	x_4	x_1
偏F	1180.16	43.04	34.57	4.72
p-value	<0.0001	<0.0001	<0.0001	<0.0001
		$F(x_2 x_3)$	$F(x_4 x_2, x_3)$	$F(x_1 x_2, x_3, x_4)$



Assignment (not required)

- ➡ 5.8使用SAS软件对习题5.5中的数据做逐步判别（取 $\alpha=0.05$ ）